Fields and Waves I

Lecture 6

Impedance Matching & Smith Charts

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These slides were prepared through the work of the following people:

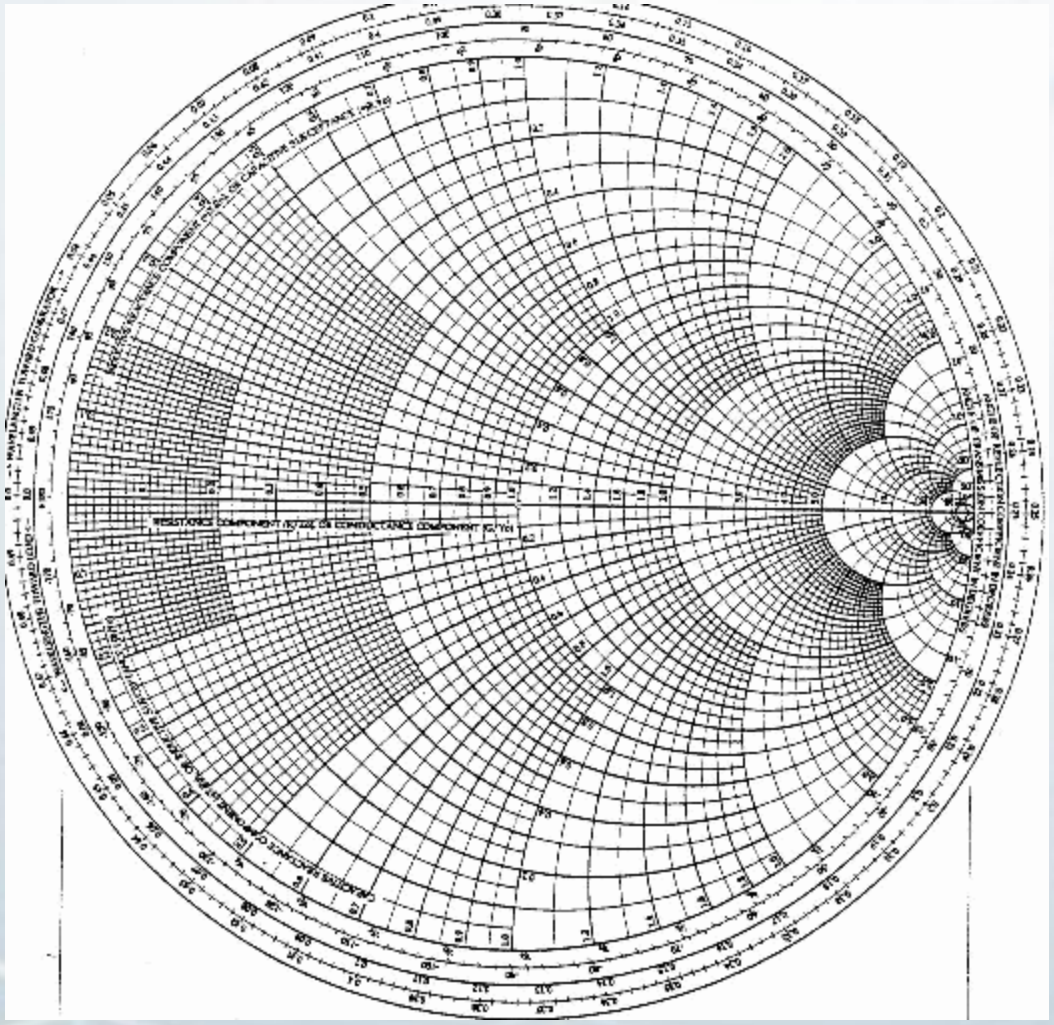
- Kenneth A. Connor – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- J. Darryl Michael – GE Global Research Center, Niskayuna, NY
- Thomas P. Crowley – National Institute of Standards and Technology, Boulder, CO
- Sheppard J. Salon – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- Lale Ergene – ITU Informatics Institute, Istanbul, Turkey
- Jeffrey Braunstein – ECE Department, University at Albany
- James Lu - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
- James Dylan Rees - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
- Alex Patterson - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY

Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

- Much of the following slides were written by Prof. Nick Shuley from the University of Queensland (UQ). The books he uses the most are:
  - Fundamentals of Engineering Electromagnetics (Cheng)
  - Fundamentals of Applied Electromagnetics (Ulaby)
  - Microwave Engineering (David Pozar)

Overview

- Impedance Matching
- Smith Charts



Review

Power on Lossless T-Lines

- General definition: $P(d,t) = v(d,t)i(d,t)$

- Instantaneous incident power:

$$P_i(d,t) = \frac{|V_0^+|^2}{Z_0} cos^2(\omega t + \beta d + \phi^+)$$

- Instantaneous reflected power:

$$P_r(d,t) = -|\Gamma|^2 \frac{|V_0^+|^2}{Z_0} cos^2(\omega t + \beta d + \phi^+ + \theta^r)$$

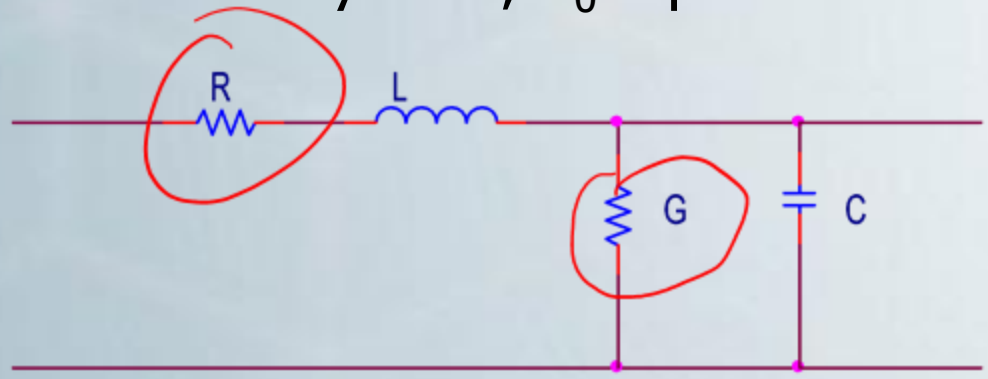
- Time-average net power:

$$P_{av} = P_{av}^i + P_{av}^r = \frac{|V_0^+|^2}{2Z_0}[1 - |\Gamma|^2]$$

Review

Lossy T-Lines

- For a lossless system, Z_0 represents



$$\frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-} \rightarrow Z_0 = \sqrt{\frac{j\omega l}{j\omega c}}$$

- To get the result for **lossy t-lines**:

Replace ***jωl*** with ***r + jωl***

Replace ***jωc*** with ***g + jωc***

Characteristic Impedance

$$Z_0 = \sqrt{\frac{r + j\omega l}{g + j\omega c}}$$

Review

Lossy T-Lines

- For lossless systems: $\beta = \omega\sqrt{lc}$
- For lossy systems: $\gamma = \alpha + j\beta = \sqrt{(r + j\omega l)(g + j\omega c)}$
- The phasors now have the factor:

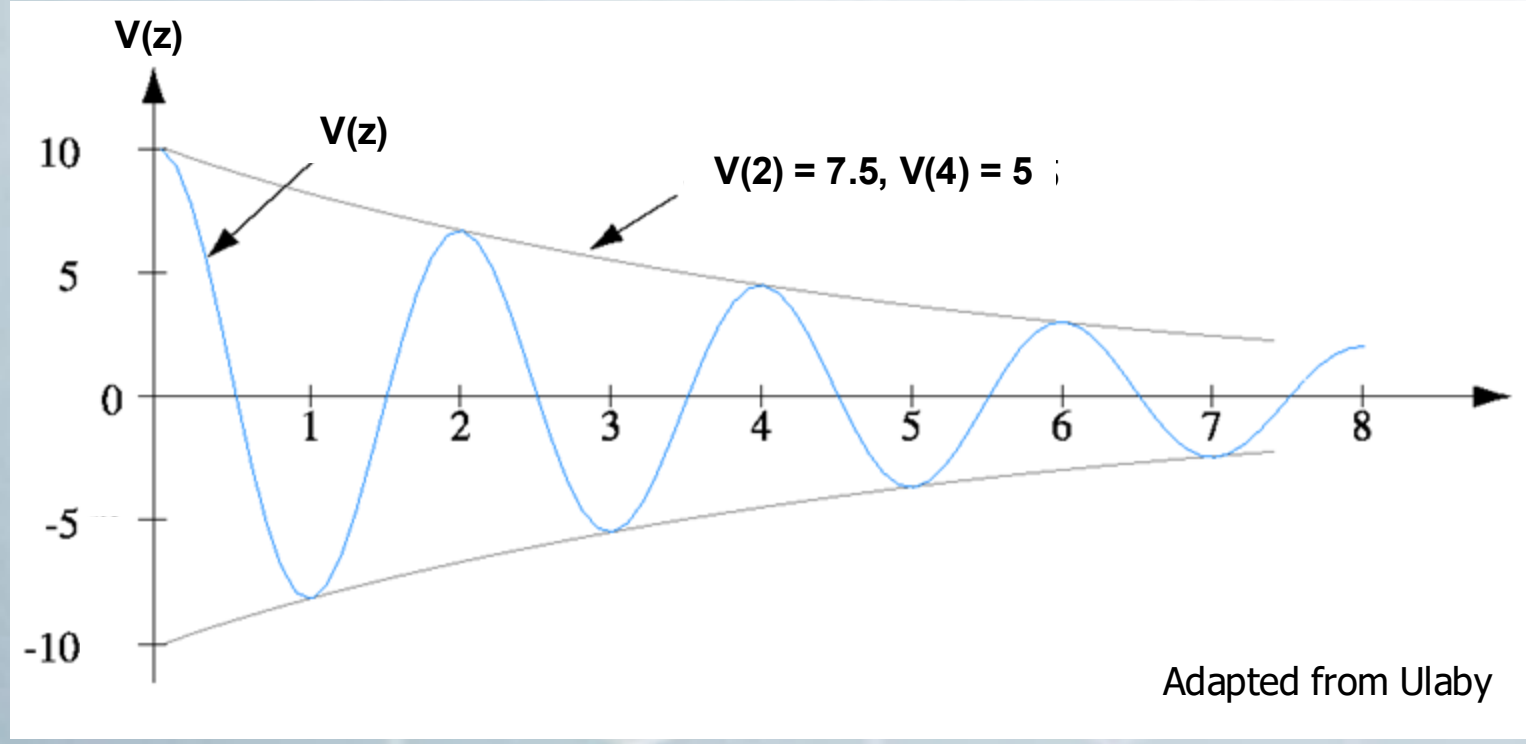
$$e^{-\gamma z} = e^{-\alpha z} \cdot e^{-j\beta z}$$

decay sinusoidal

Review

Lossy T-Lines

- The attenuation factor α is due to ohmic losses on the line.
- Looking at a plot like the one below, how do you find α and β?



Review

Lossy T-Lines

- If r is small compared to ωL (and $g = 0$): we have a low-loss t-line
- Low-loss lines have the following properties:

$$\alpha = r\sqrt{\frac{c}{l}}$$

$$\beta = \omega\sqrt{lc} = \frac{\omega}{u}$$

$$Z_0 = \sqrt{\frac{l}{c}}$$

Review

Distortionless T-Lines

- For practical lines, the conductance per unit length g is negligible. What can we do to eliminate the frequency dependence of Z_0?

$$\boxed{\frac{r}{l} = \frac{g}{c}}$$

This is called the **Heaviside condition**

Impedance Matching

Impedance Matching

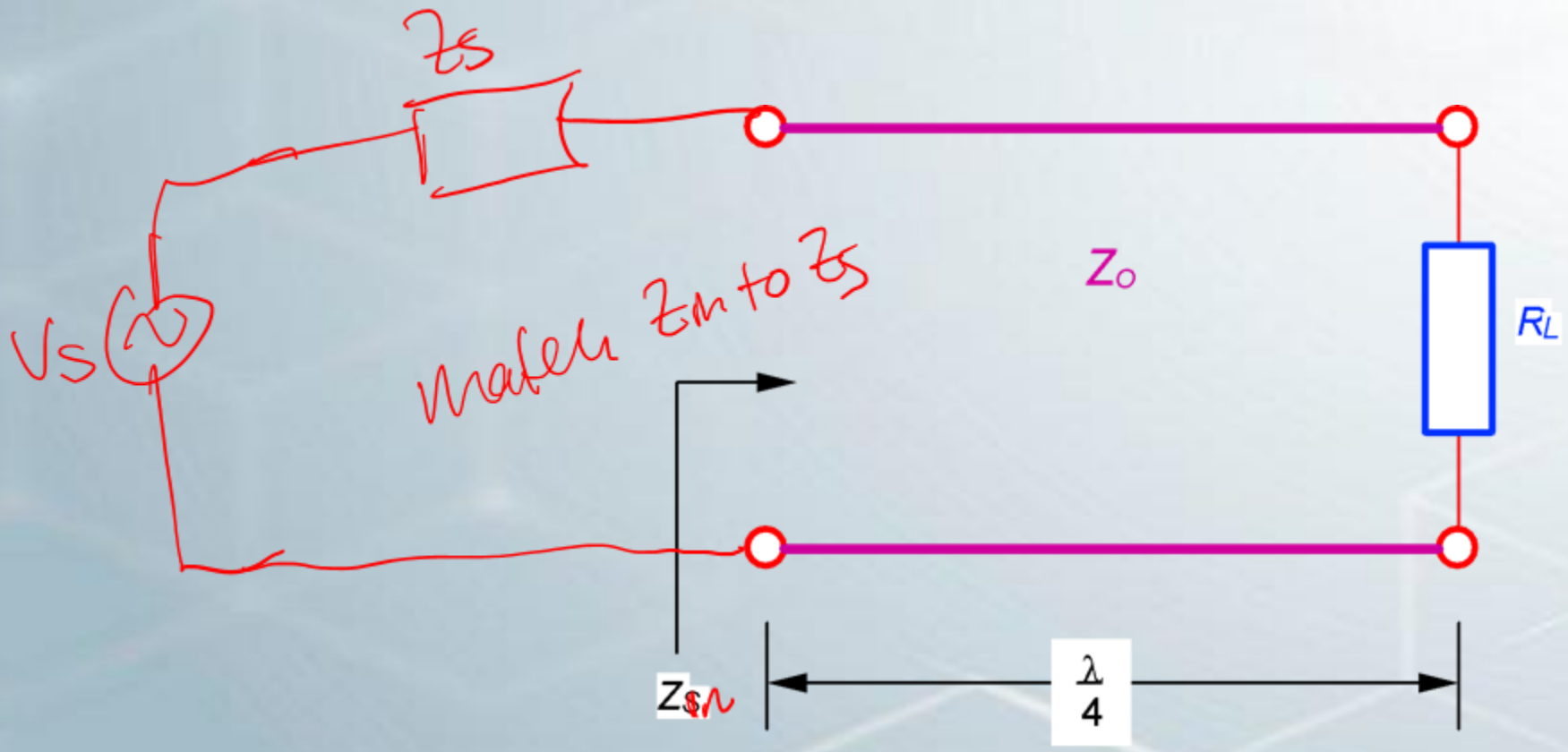
- For a transmission line to work well, we must not only engineer the line parameters properly but also match it to its load.
- Reflections lead to variations in the input impedance of the line. The input impedance changes with line length and frequency.
- Power is wasted. An impedance match provides maximum power transfer to the load.
- A VSWR > 1 means there will be voltage maxima on the line. These can lead to voltage breakdown at high power levels.

Impedance Matching

- If we match the load to Z_0, the input impedance remains constant at the value Z_0 (no reflection).
- In this case, the input impedance is independent of line length and frequency (over the bandwidth of the matching network).
- VSWR = 1. Therefore there are no voltage peaks on the line.
- Maximum power transfer to the load is achieved.

Impedance Matching

- Consider a lossless quarter-wavelength of line terminated by a resistance R_L:



Impedance Matching

Assuming the line is lossless: $Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$

and $\tan \beta l = \tan\left(\dfrac{2\pi}{\lambda}\dfrac{\lambda}{4}\right) = \tan\left(\dfrac{\pi}{2}\right) = \infty$

so $Z_0 \tan \beta l = Z_0 \dfrac{jZ_0}{jZ_L} = \dfrac{Z_0^2}{R_L} = Z_{in}$

choose Z_0

Note that Z_{in} is purely real, so the $\lambda/4$ line allows us to transform one resistance value to another resistance.

We can exploit this to impedance match.

Impedance Matching

Key Properties of λ/4 Lines

- **Normalized impedance** is a way to express impedances as a ratio with the characteristic impedance.

$$z = \frac{Z_L}{Z_0} = \frac{|Z_L|e^{j\theta_L}}{|Z_0|e^{j\theta_0}} = \frac{|Z_L|}{|Z_0|}e^{j(\theta_L-\theta_0)} \quad \text{if } Z_0 \text{ real: } z = \frac{R_L}{Z_0} + j\frac{X_L}{Z_0}$$

- In normalized terms, a matched impedance would be 1. (This will be useful soon.)
  - If Z_0 = 25Ω , what is the impedance 50+j75 Ω expressed in normalized terms?

$2 + j3$

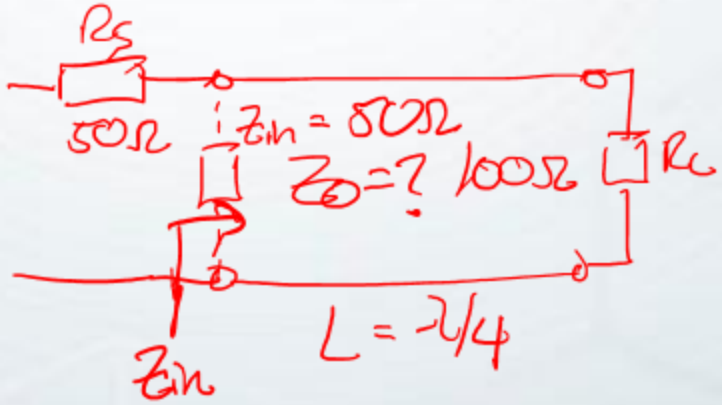
Impedance Matching

Key Properties of λ/4 Lines

- A mismatched load can be matched to a t-line line using a $\lambda/4$ transformer of suitable characteristic impedance.
- Example: match a 100 Ω resistor to a 50 Ω line.

$$R_L = 100\ \Omega$$

$$R_S = 50\ \Omega$$



∴ the transformer characteristic impedance Z_{OT} must be:

match Zin to Rs

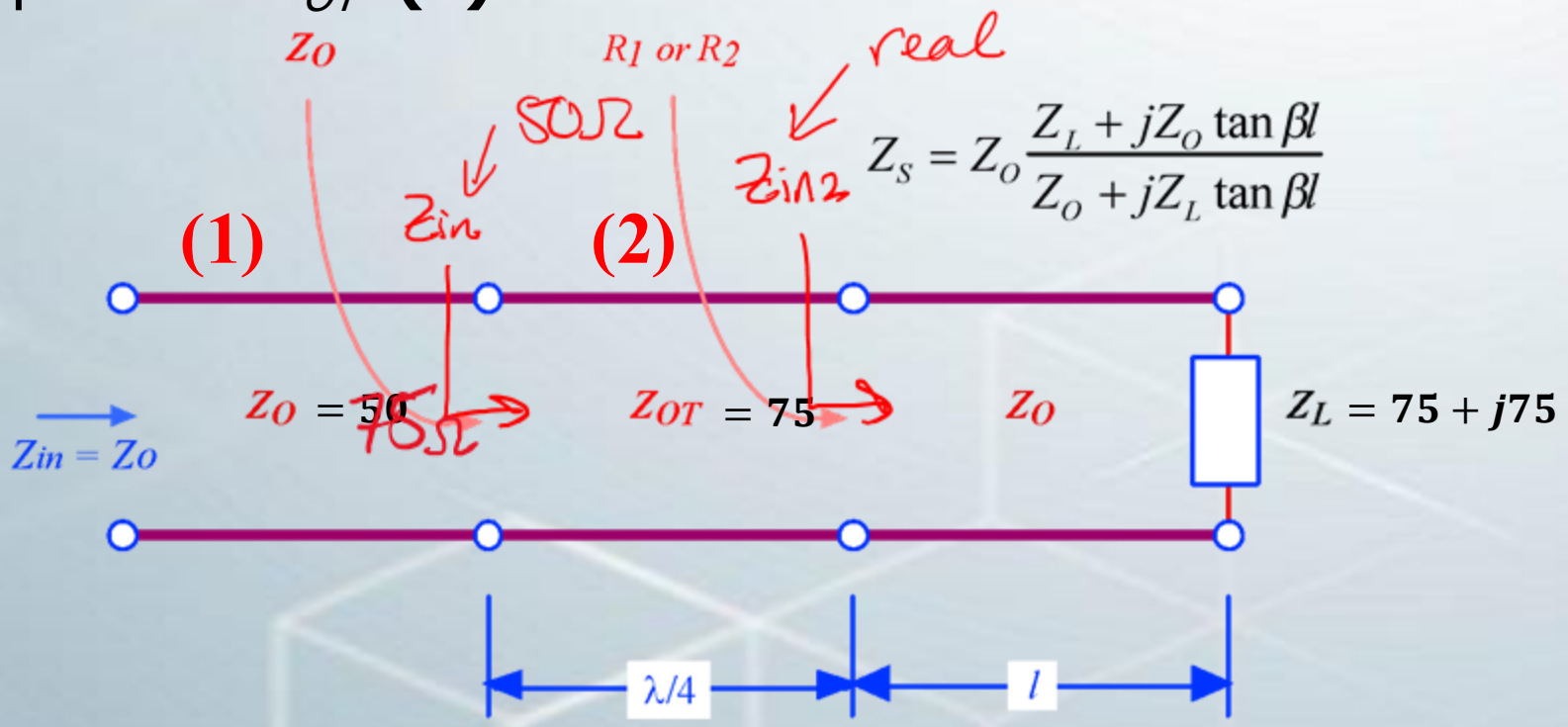
$$Z_{in} = \frac{Z_0^2}{Z_L} \rightarrow Z_0 = \sqrt{Z_{in} \cdot Z_L} = \sqrt{50\Omega \cdot 100\Omega} = \underline{70.7\Omega}$$

(Z_{in}: 50Ω; Z_L: 100Ω)

Impedance Matching

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$$

- If Z_L is not real, a length of line (with characteristic impedance Z_O) may be used to transform Z_L to a real impedance **(2)**
- That can then be converted to Z_O by the quarter-wave transformer, of characteristic impedance Z_{OT}. **(1)**



Impedance Matching

- How do we find a length of line that makes the input impedance of that first segment real?
- We know that the input impedance is periodic in λ/2 with line length. But the math is tedious.
- What if we had a system where we could map a load impedance onto a circle, rotate it by βL like we rotate a clock, and read out the input impedance?
- There's a chart that does this, and it's called a Smith Chart.



Impedance Matching

- Invented by American engineer Philip Hagar Smith in 1939 for antenna input impedance matching
- Also invented in parallel by Japanese engineer Mizuhashi Tosaku in 1937 (the chart is also called the Mizuhashi Chart or Smith-Mizuhashi Chart)
- Allows us to graphically solve the relation between reflection coefficients and load impedance and/or find input impedance, saving us time.



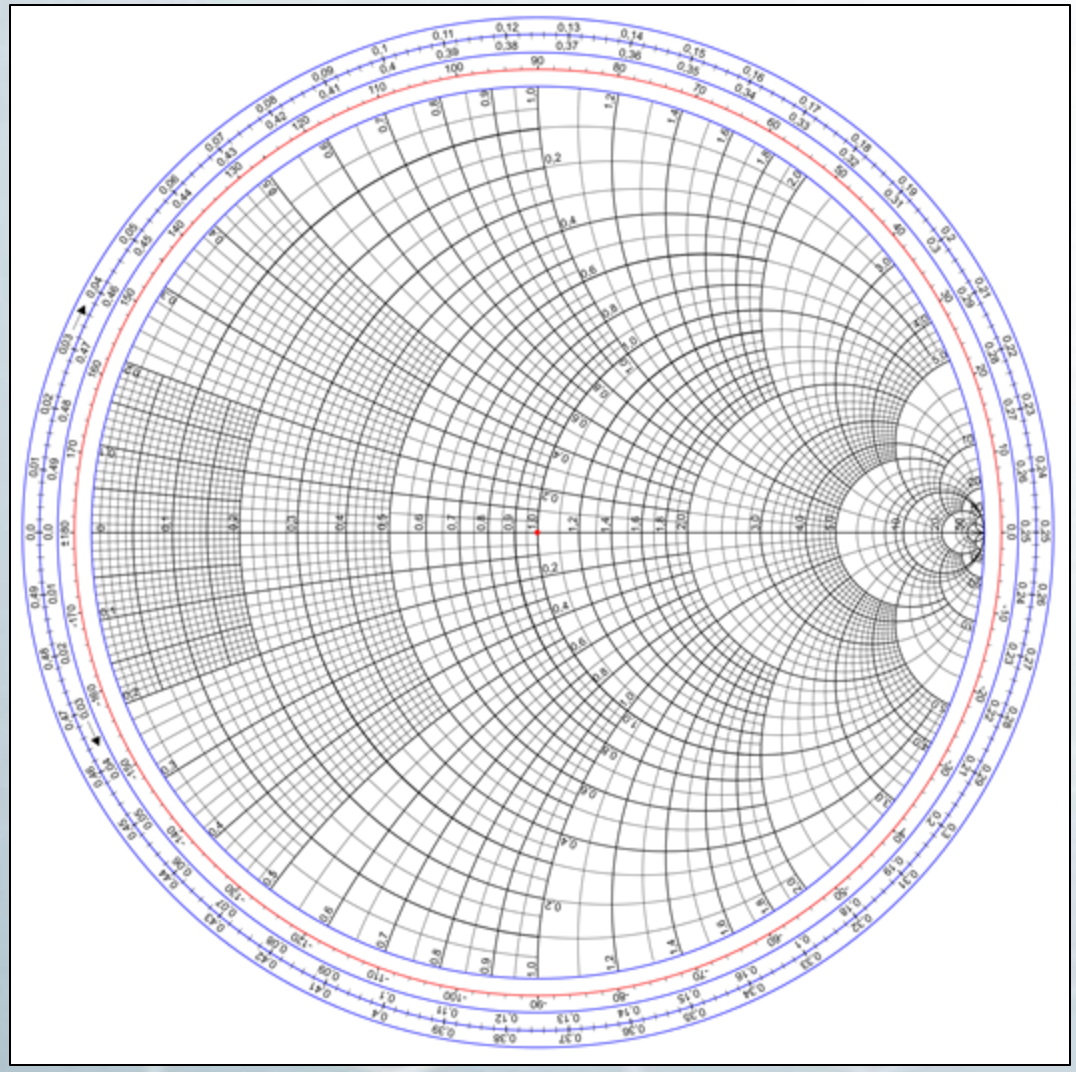
Philip H. Smith

Alternative: solve this eq. for l every time

Solve for l

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$$

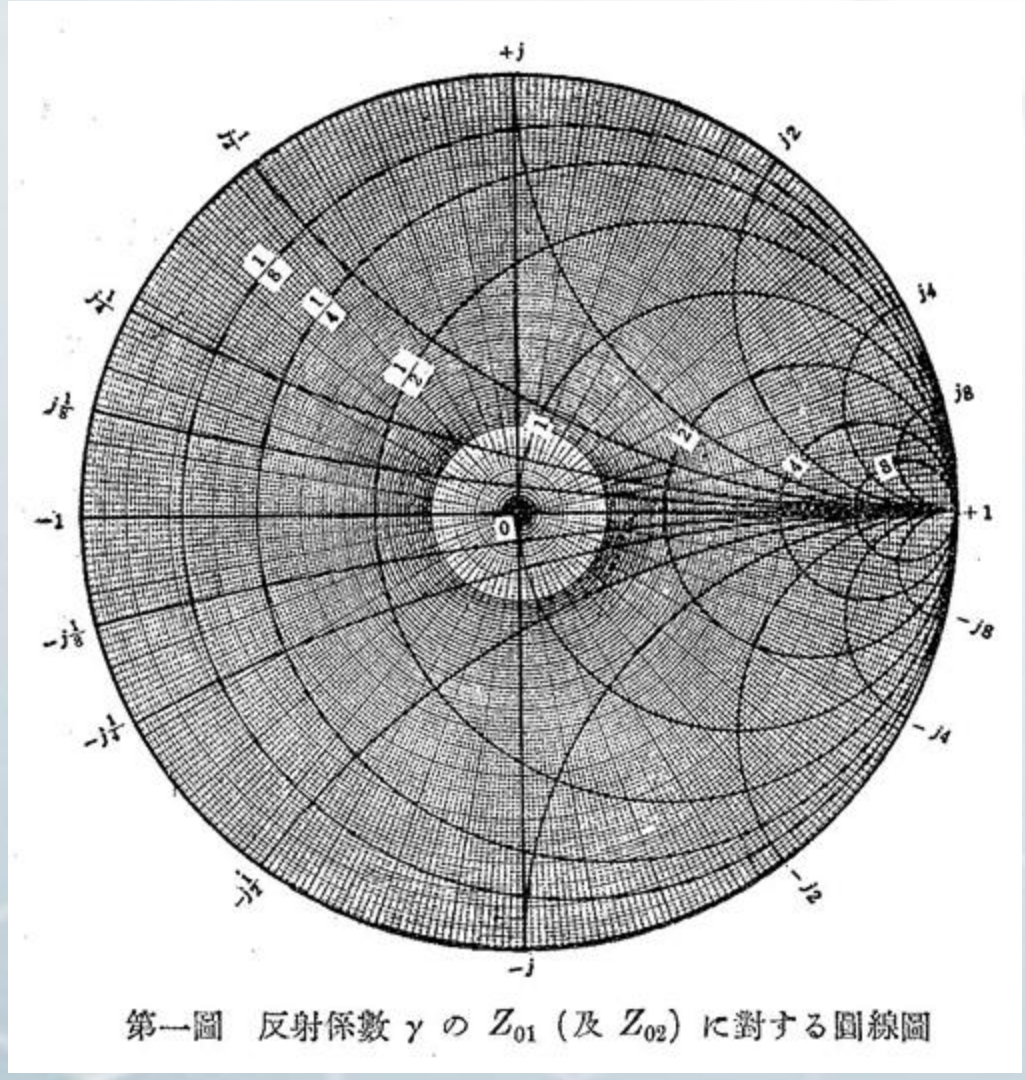
Impedance Matching



Mizuhashi Chart

Source: Wayback Machine

Note that both US and Japanese versions use the same curved coordinate system



第一圖　反射係數 γ の Z_{01}（及 Z_{02}）に對する圖線圖

Impedance Matching

- Recall: reflection coefficient is a complex quantity in general:

$$\Gamma = |\Gamma| e^{j\theta_r} = \Gamma_r + j\Gamma_i \qquad \ldots(6.1)$$

- We can thus plot reflection coefficients in the complex Γ plane.
- The components are:

$$\Gamma_r = |\Gamma| \cos\theta_r$$
$$\Gamma_i = |\Gamma| \sin\theta_r$$

Smith Chart - Complex Γ-plane

$$z = r + jx$$

$$\Gamma = \frac{z-1}{z+1}$$

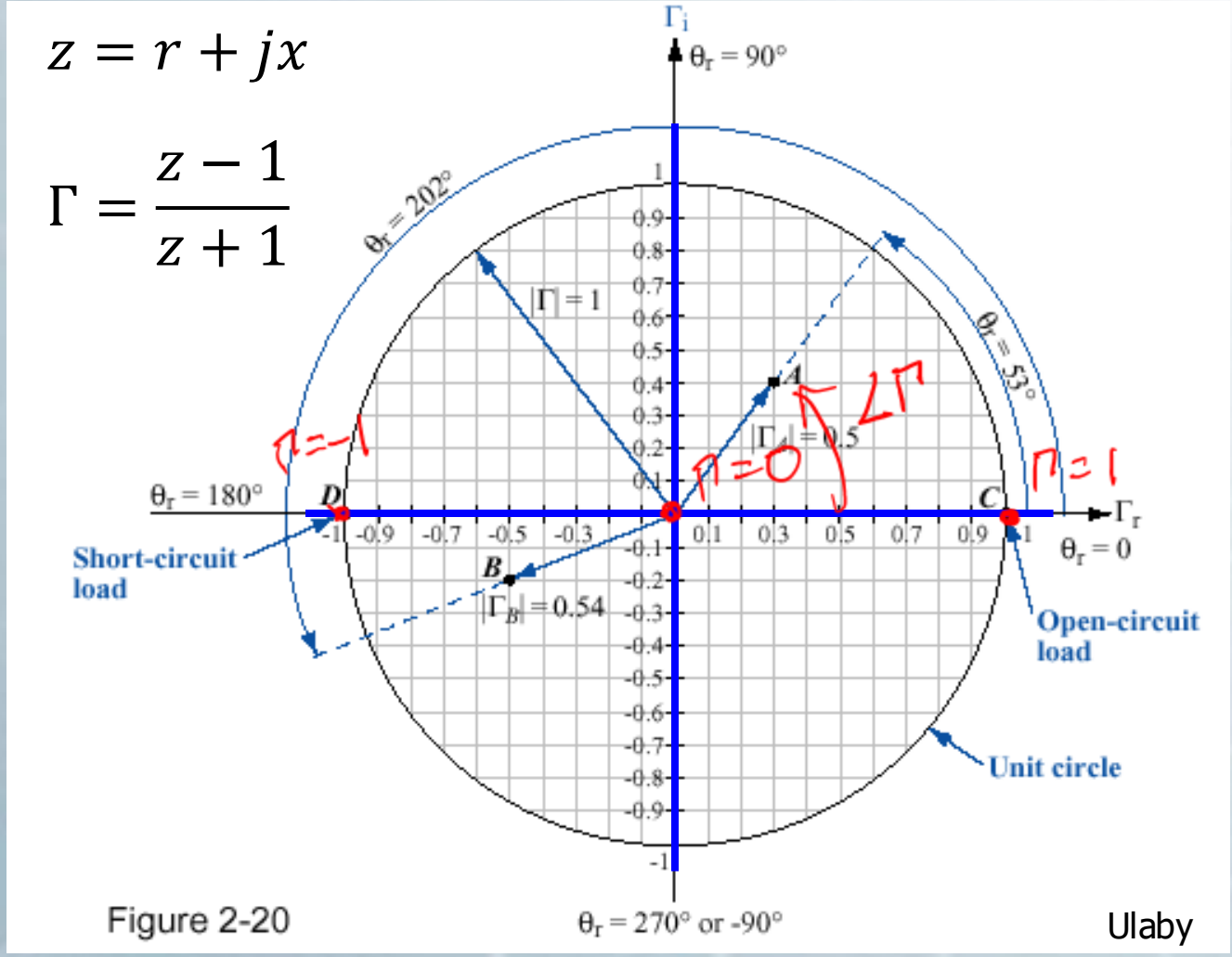


Figure 2-20 Ulaby

- For Z_0 real:

$$z = \frac{R_L}{Z_0} + j\frac{X_L}{Z_0}$$

Let's orient ourselves in the complex Γ plane

Smith Chart

- We need to relate impedances to reflection coefficients:

- First, we normalize all impedances with respect to the characteristic impedance of the line:

$$z = \frac{Z}{Z_0} \qquad \text{e.g.} \quad z_L = \frac{Z_L}{Z_0}$$

- For some impedance Z_R, expressing in terms of Γ :

$$\Gamma = \frac{Z_R - Z_0}{Z_R + Z_0} = \frac{Z_R/Z_0 - 1}{Z_R/Z_0 + 1} = \frac{z_R - 1}{z_R + 1} \Leftrightarrow z_R = \frac{1+\Gamma}{1-\Gamma} \quad ..(6.2)$$

Smith Chart

- Now since the normalized impedance can be written as:

$$z_R = r_R + jx_R \qquad ..(6.3)$$

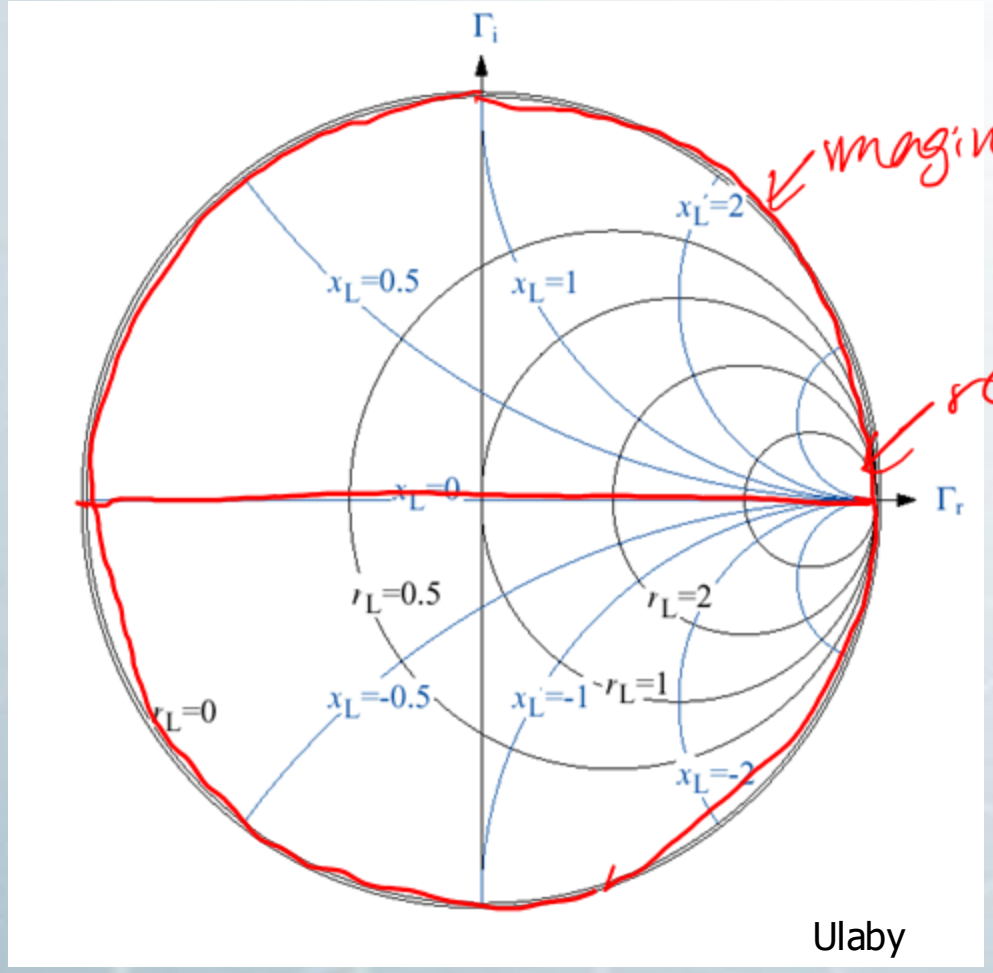
- we set (6.3) equal to (6.2) using the real and imaginary parts of (6.1). This gives:

$$r_R + jx_R = \frac{1+\Gamma}{1-\Gamma}$$

- We can then solve for the r_R and x_R in terms of Γ. Graphical families of all possible solutions to this equation constitute the Smith Chart.

Smith Chart – Normalized Impedance

A Smith chart is therefore a polar plot of Γ, with contours of real and imaginary parts of z superimposed on top.



This allows easy conversion between normalized impedance z and Γ

$$r_L + jx_L = \frac{1+\Gamma}{1-\Gamma}$$

- Where is $z_L = 1 + j$?
- Where is $z_L = 1 - j$?
- Where is $z_L = -j0.5$?

Smith Chart

- A Smith chart printout is available on LMS (Smith Charts)
- This chart is designed to be printed out and done by hand with a ruler
- However, you can also use it on your computer using digital tools

Smith Chart

- Example:
  - Let's say that a t-line has a characteristic impedance of 40Ω and a load impedance of 20+40jΩ.
  - What is the magnitude and phase of the reflection coefficient?

1) calculate the normalized impedance: $z_L = \frac{Z_L}{Z_0} = 0.5 + j$

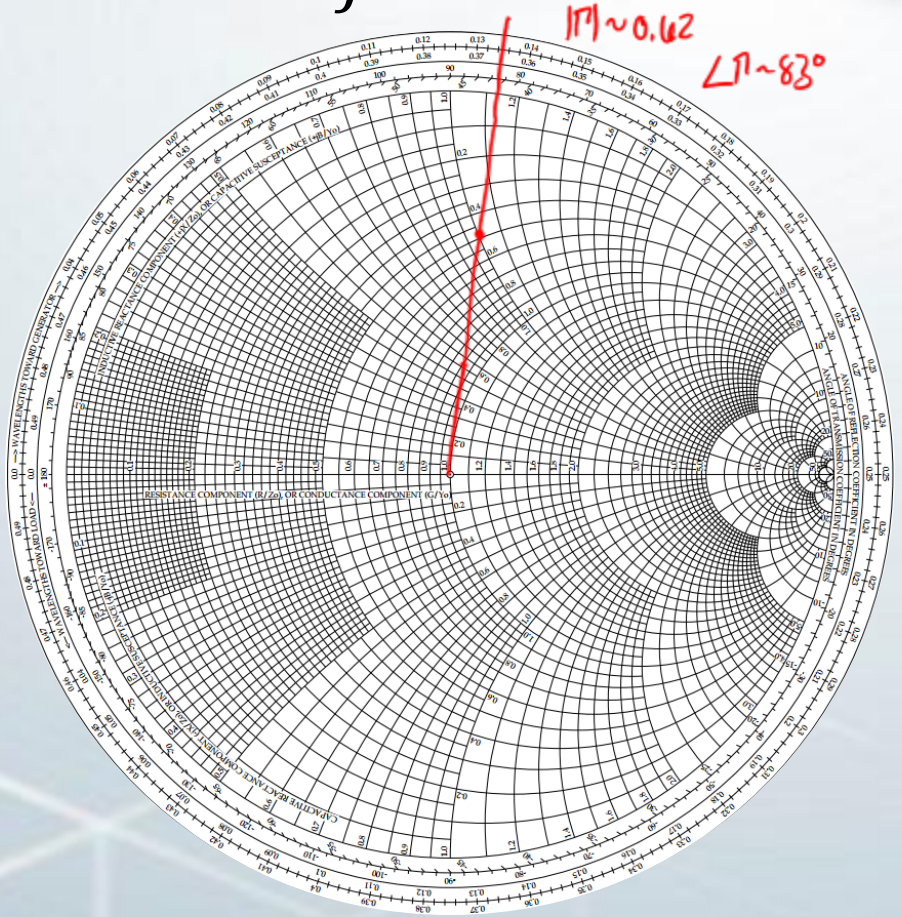
2) $\Gamma = \frac{z-1}{z+1} = 0.0769 + j0.615 = 0.62 \angle 82.9^\circ$ (numerical version)

(smith chart version: see next slide)

Smith Chart

- The reflection coefficient is proportional to the length of the radial vector on the chart.
- The length of the vector to the periphery corresponds to $|\Gamma| = 1$
- The phase angle of the reflection coefficient is measured from the positive direction of the horizontal axis

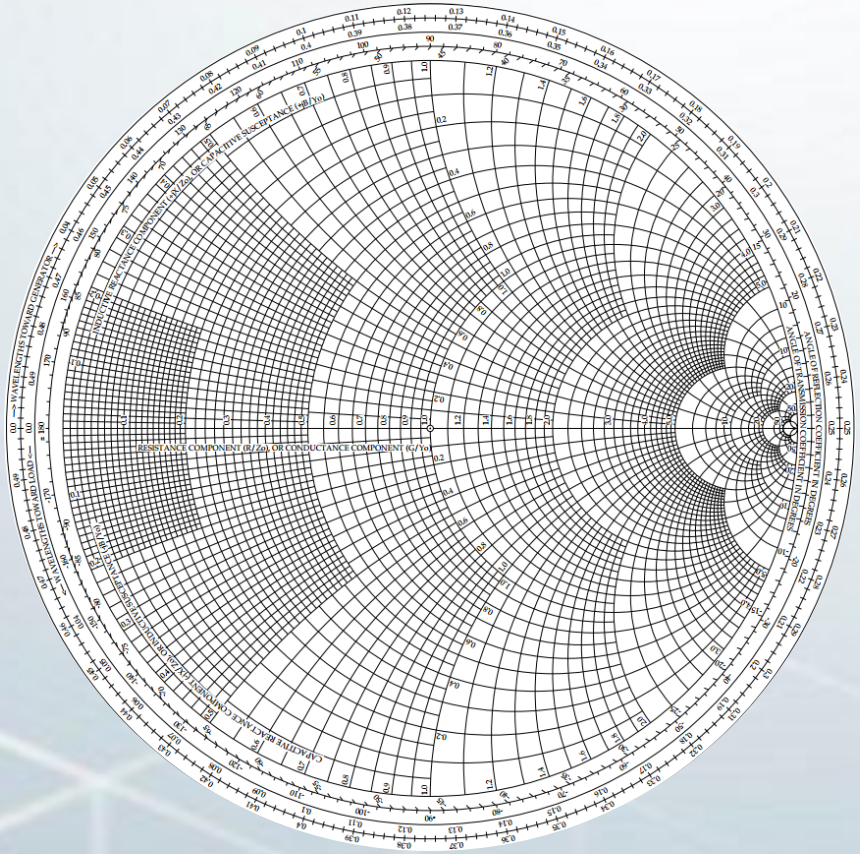
$$Z = 0.5 + j$$



Smith Chart

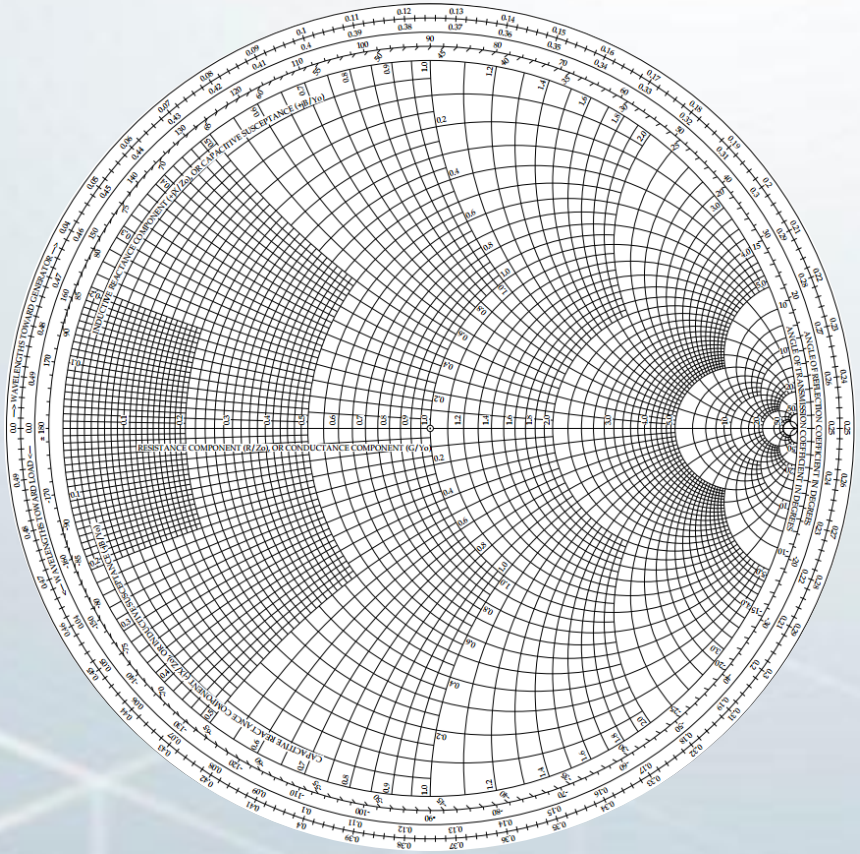
All impedances in the top half are inductive e.g. 1+j

All impedances in the bottom half are capacitive e.g. 1-2j



Smith Chart

purely real impedances are along the horizontal center line



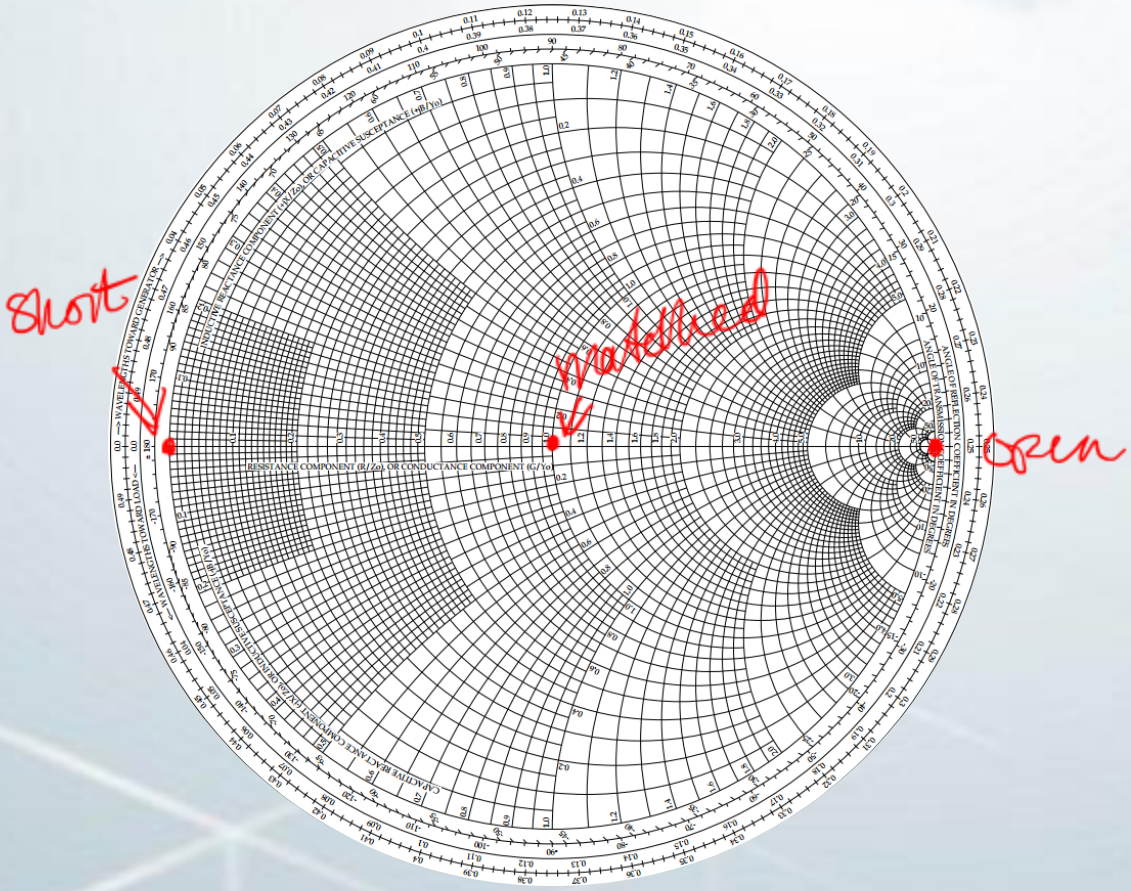
purely imaginary impedances are along the periphery

Smith Chart

open circuit point
(infinite impedance)

unity impedance z =1
(match point)

short circuit point
(zero impedance)



Smith Chart

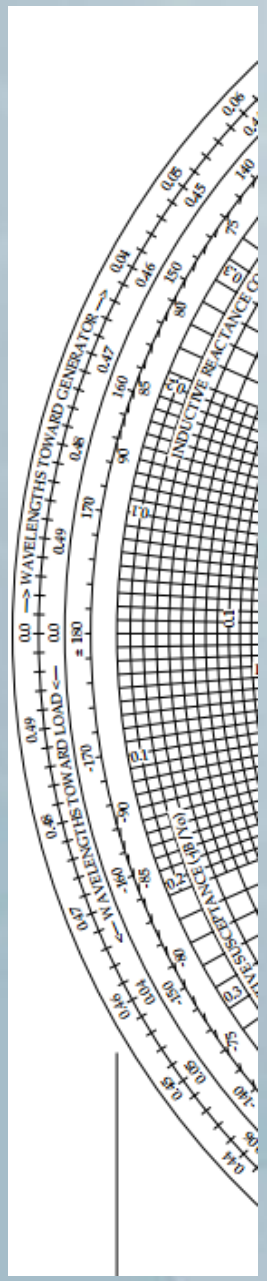
- What do you notice about the angle between the open and short circuit on the Smith Chart?

180°

- What length of transmission line is required to make an open circuit look like a short circuit or vice versa?

$L = \lambda/4$

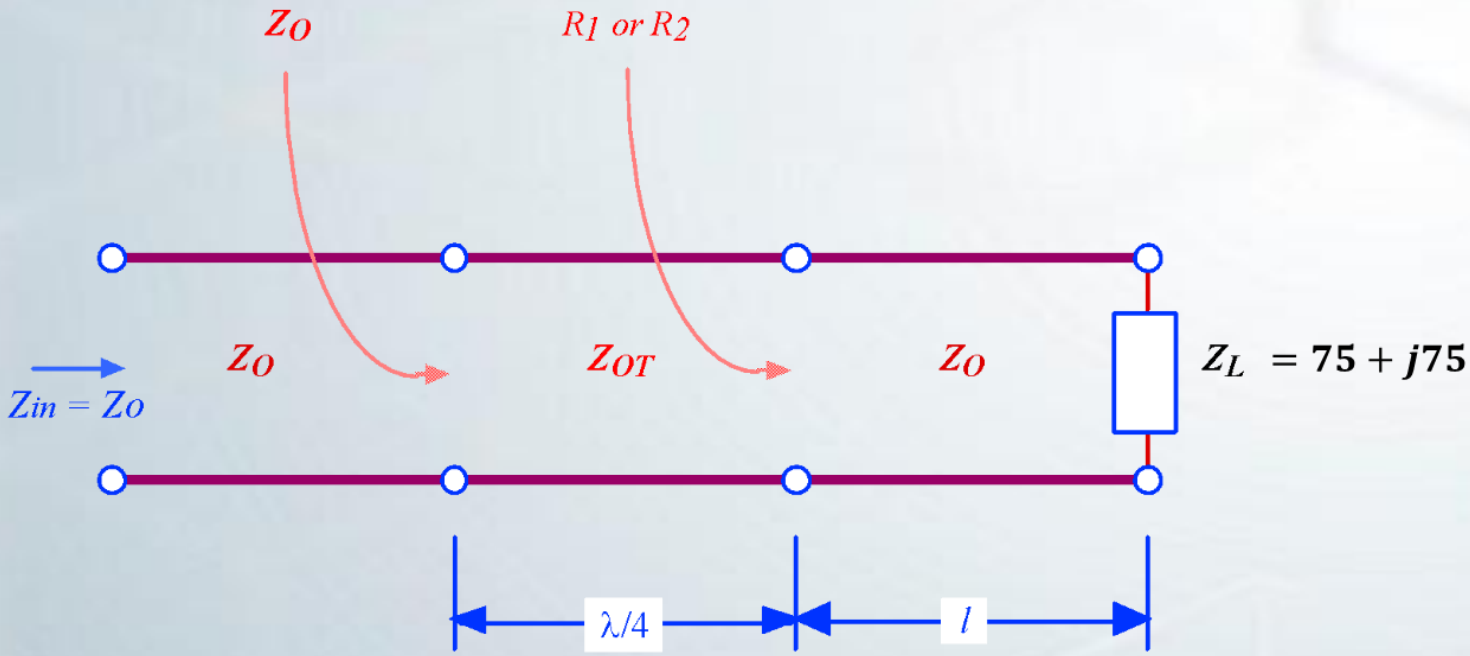
- *Thus, the angle on a Smith chart is also measured in wavelength (of the AC input signal).*



Smith Chart

- We can use the Smith chart to find the length of line L required to give a certain input impedance.
- We **start at the normalized load impedance**, travel along a circle **towards the generator**, and stop at the input impedance we want to provide
- Where we stop is the **input impedance of the t-line of length L**. But how do we read off this length?
- There are two scales on the periphery (in wavelengths)
  - 1 **towards generator** (clockwise)
  - 1 towards load (counterclockwise)
- Read off the distance (in wavelengths) you traveled from the initial impedance to the final impedance. This is the line length L.
- Note that once around the whole chart is a total length of λ/2

Smith Chart

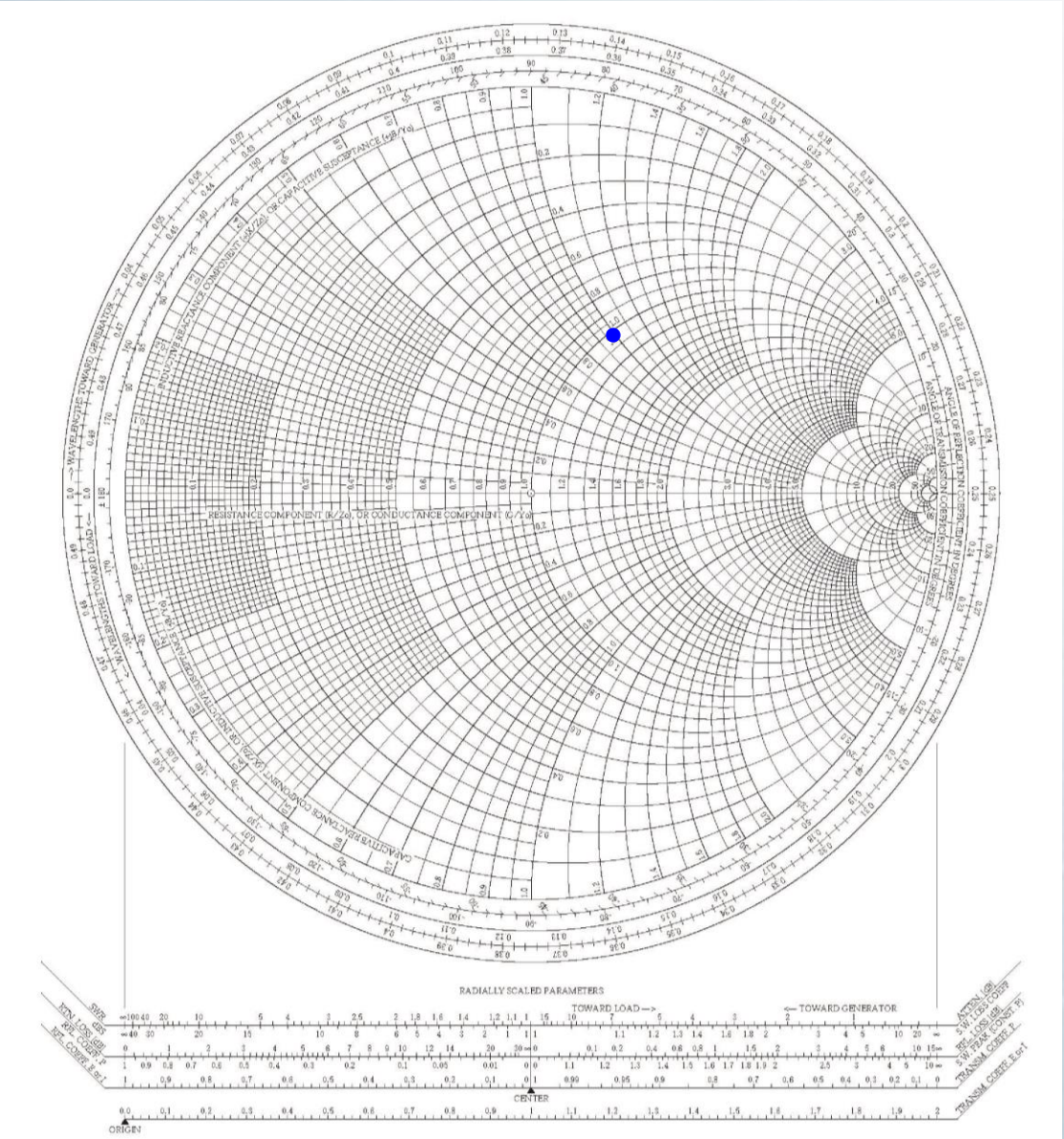


- What length of line l presents a **real** input impedance, when $Z_0 = 75\Omega$ and $Z_L = 75 + j75\ \Omega$?

1. Plot the normalized load impedance z_L on the Smith chart

$z_L = 1 + j$

Smith Chart

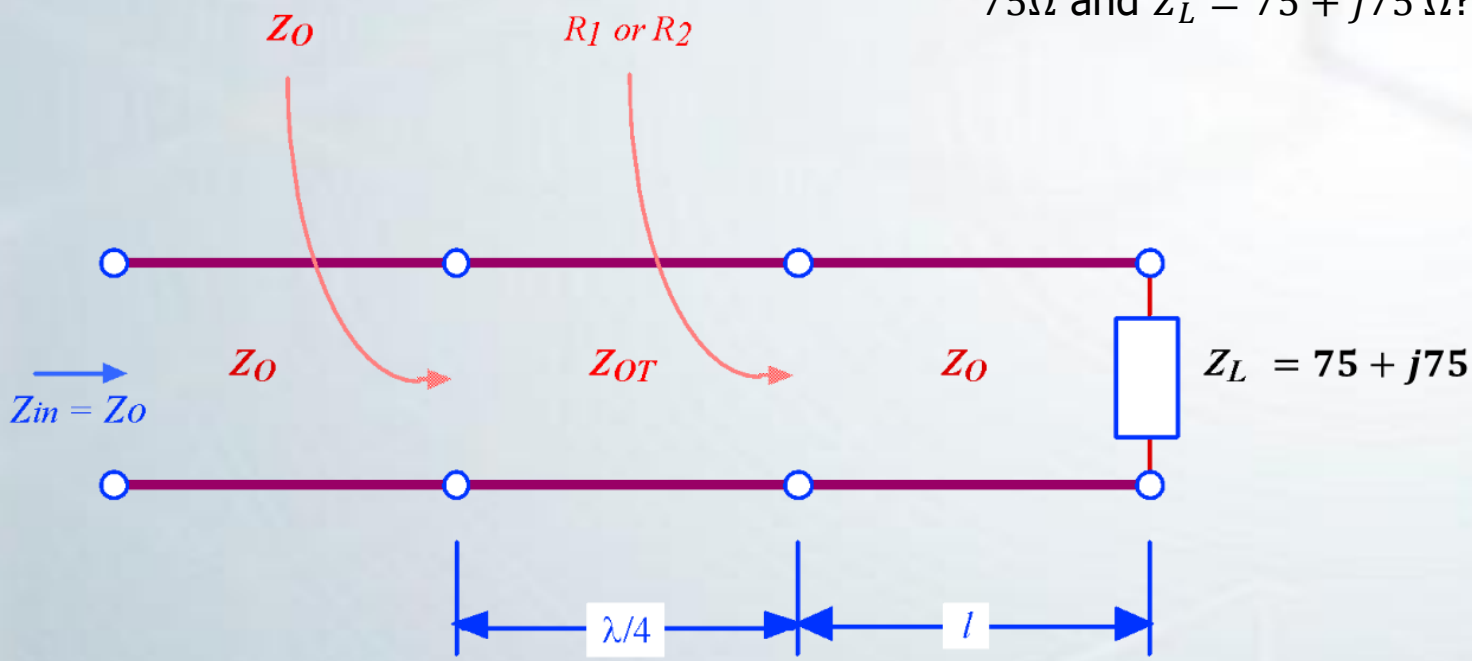


1. Plot the normalized load impedance z_L on the Smith chart

$Ze = 1+j$

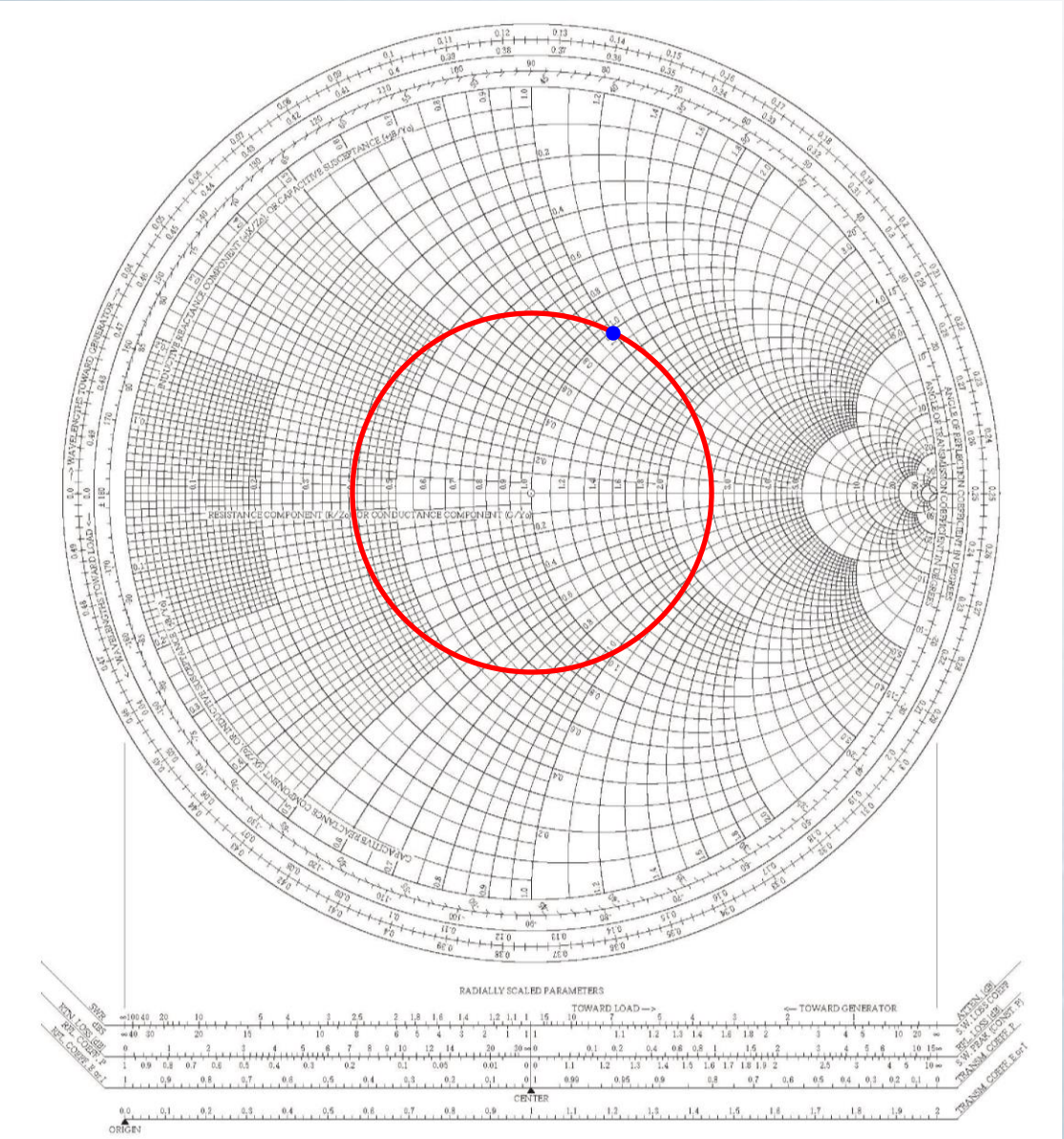
Smith Chart

- What length of line l presents a **real** input impedance, when $Z_0 = 75\Omega$ and $Z_L = 75 + j75\ \Omega$?



1. Plot the normalized load impedance z_L on the Smith chart
2. Draw a circle of constant radius through the plotted impedance (centered at origin of Smith chart) – VSWR circle

Smith Chart

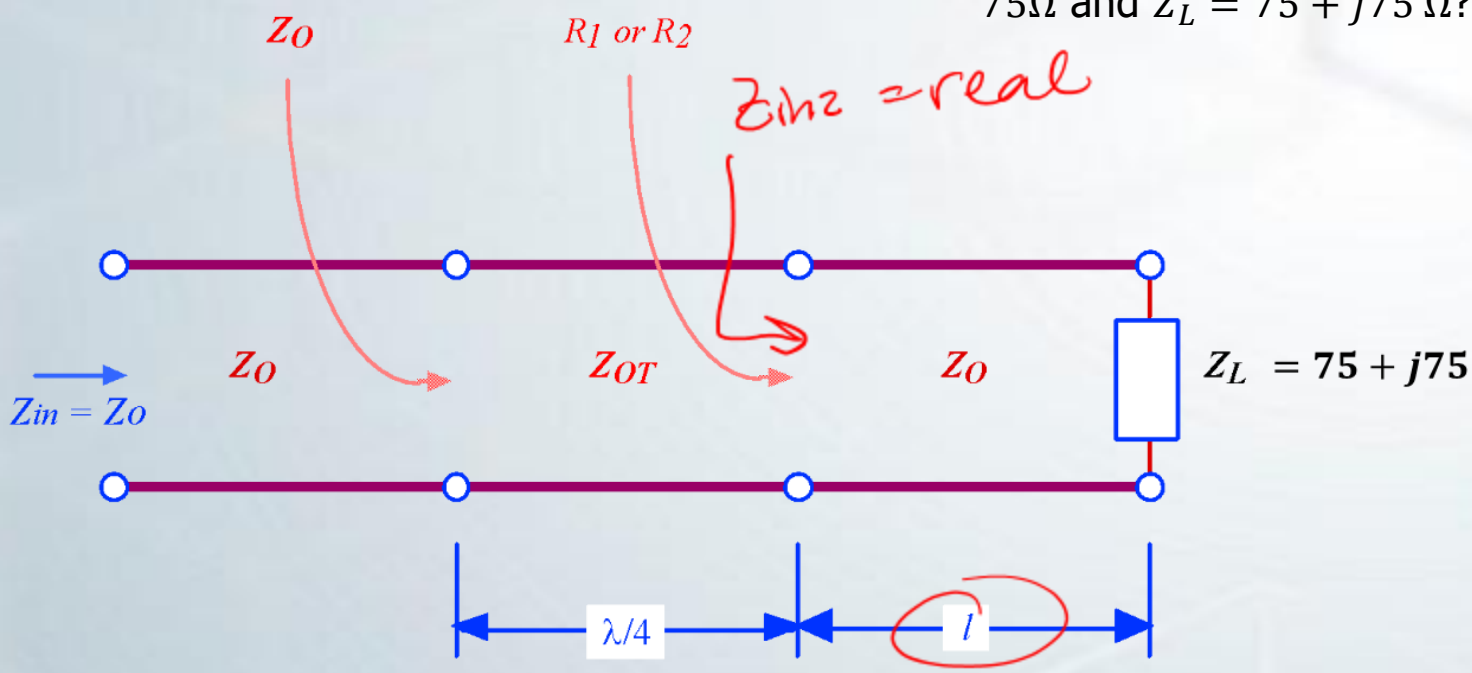


2. Draw a circle of constant radius, centered at the origin of the Smith chart, through the plotted impedance.

   This is called the **VSWR circle**

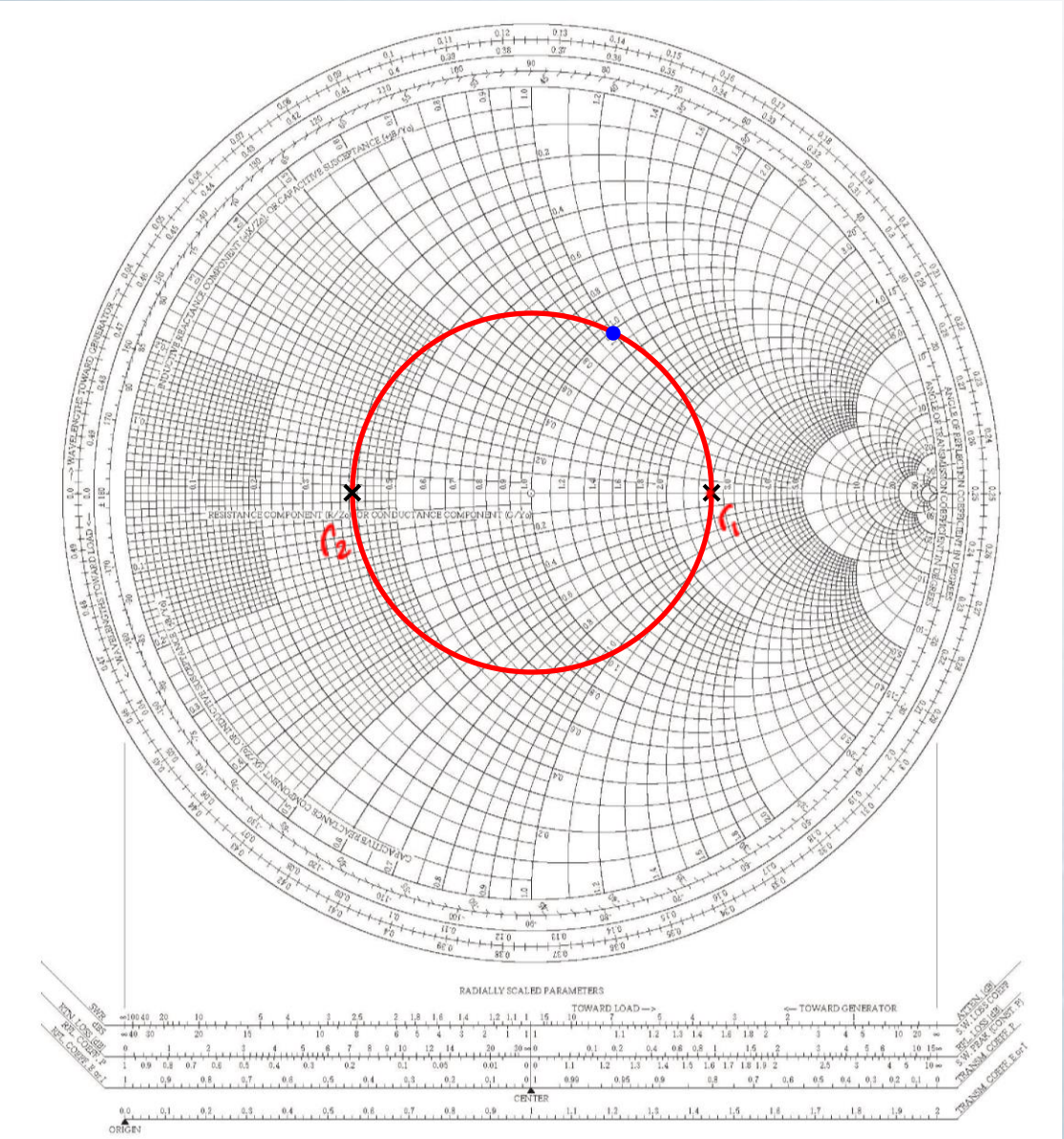
Smith Chart

- What length of line l presents a **real** input impedance, when $Z_0 = 75\Omega$ and $Z_L = 75 + j75\ \Omega$?



1. Plot the normalized load impedance z_L on the Smith chart
2. Draw a circle of constant radius through the plotted impedance (centered at origin of Smith chart) – VSWR circle
3. Find where the circle intersects the x-axis of the chart – these are all the possible real input impedances

Smith Chart



3. Find where the circle intersects the x-axis of the chart – these are all the **possible real input impedances**

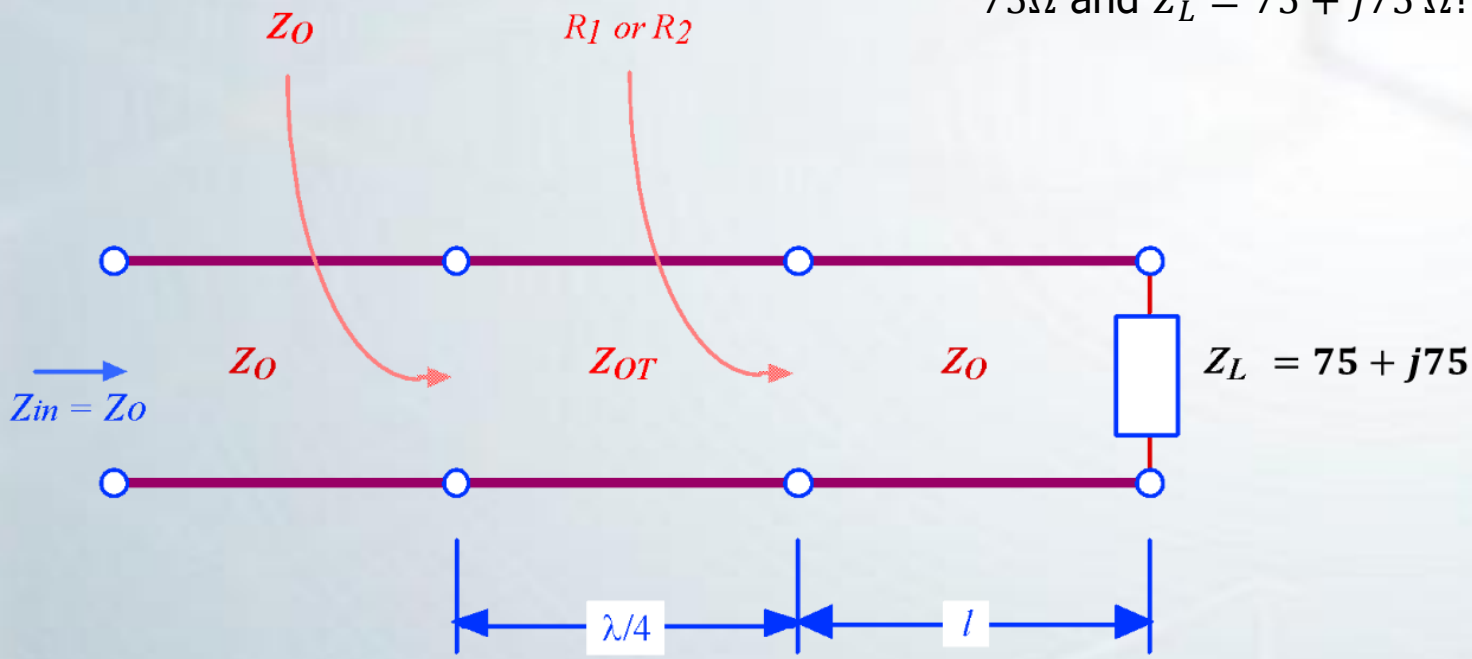
$r_1 \cong 2.6 \rightarrow R_{in1} = 2.6 \cdot 75\Omega = 195\Omega$

OR

$r_2 \cong 0.38 \rightarrow R_{in2} = 0.38 \cdot 75\Omega = 28.5\Omega$

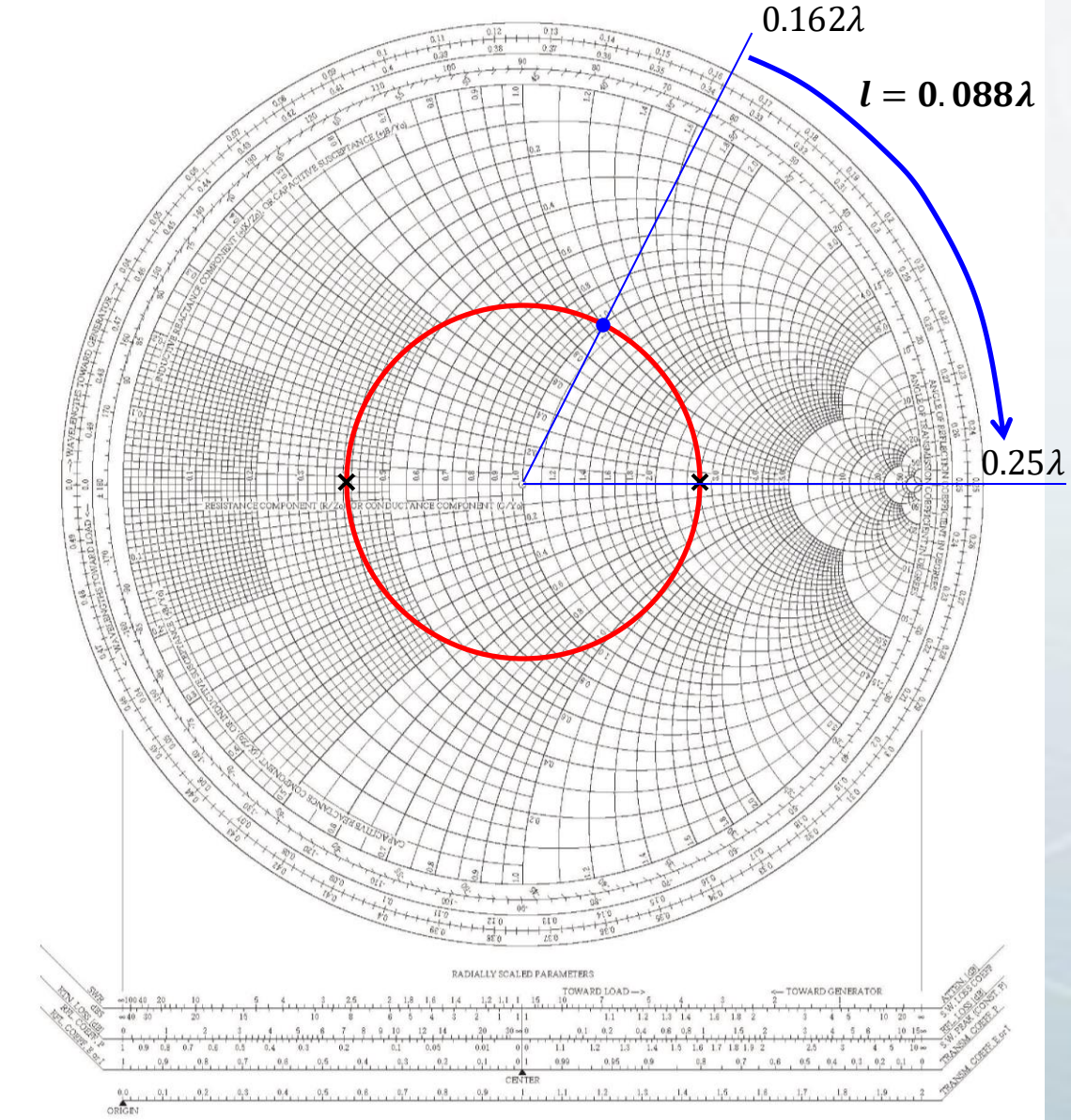
Smith Chart

- What length of line l presents a **real** input impedance, when $Z_0 = 75\Omega$ and $Z_L = 75 + j75\ \Omega$?



1. Plot the normalized load impedance z_L on the Smith chart
2. Draw a circle of constant radius through the plotted impedance (centered at origin of Smith chart) – VSWR circle
3. Find where the circle intersects the x-axis of the chart – these are all the possible real input impedances
4. Choose a real input impedance and read off the required l

Smith Chart



4. Choose a real input impedance and read off the required l

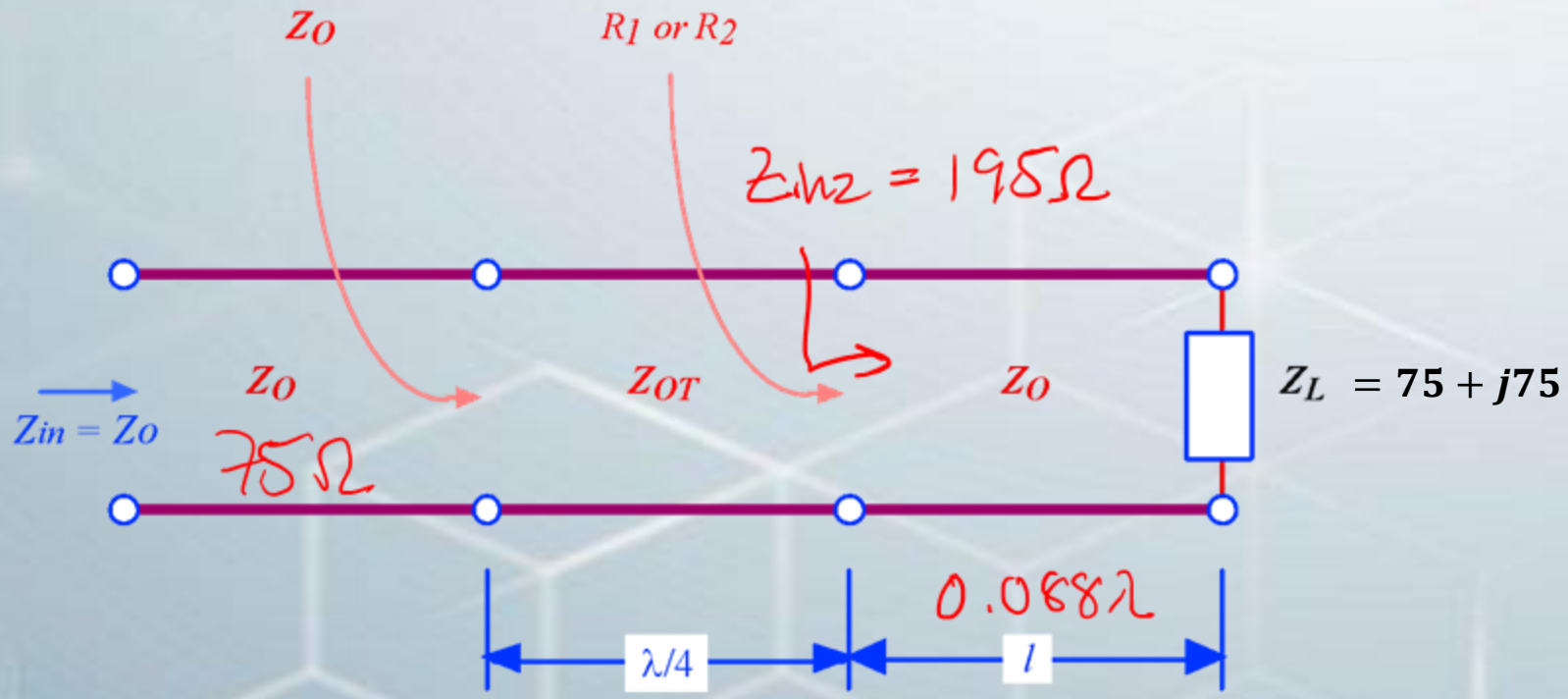
- (1) Draw a line from the origin through the load impedance to the edge of the chart. Read off the number on the "Wavelengths toward generator" scale.
- (2) Do the same for your chosen real input impedance.
- Calculate the total distance required to travel from (1) to (2)

a line of length $L = 0{,}088\lambda$
presents $Z_{in} = 195\,\Omega$

Smith Chart

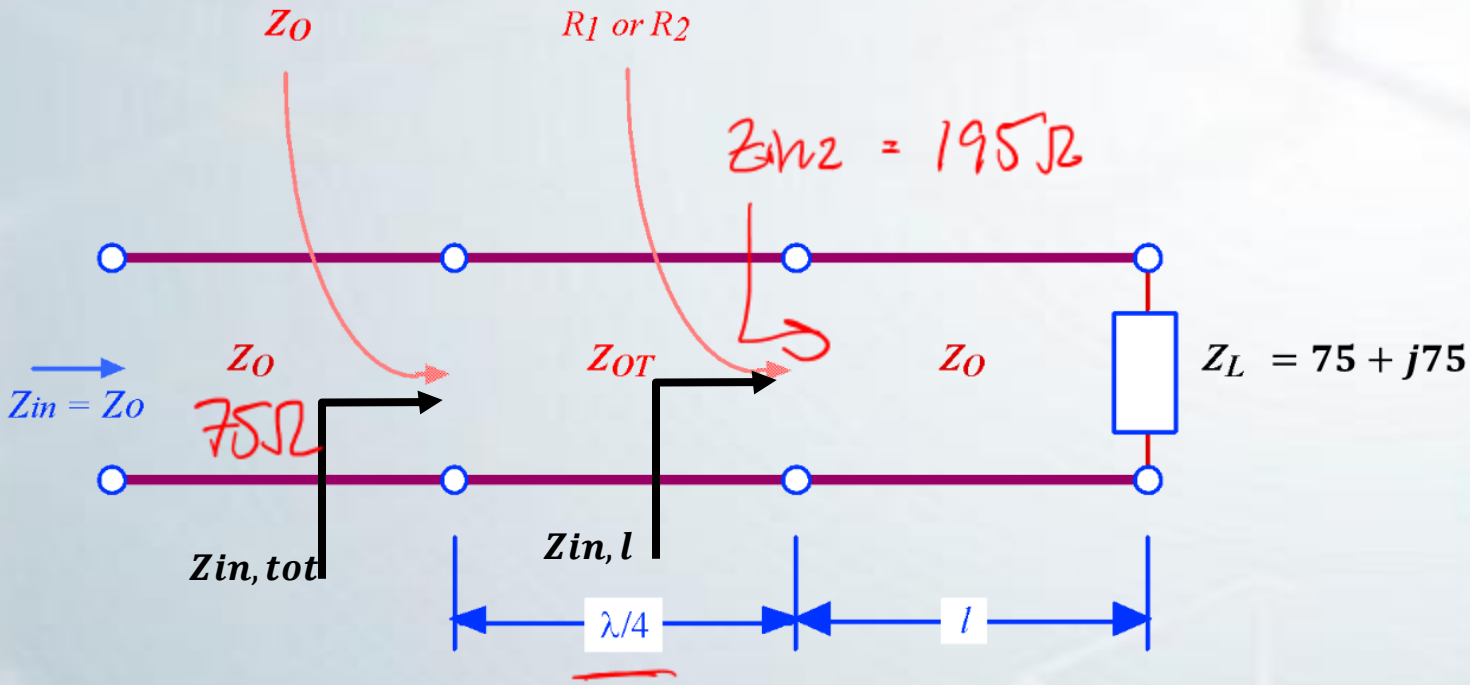
We can match a load to a transmission line using:

1. A specific length of t-line l attached to the load that makes the input impedance looking down the t-line real **DONE**
2. A $\lambda/4$ transformer attached to the input of the t-line in (1) to make the input impedance looking down the series combination equal to the impedance we want to match (Z_0 here) **NEXT**



Do Lecture 6 Exercise on Gradescope in groups of up to 4.

Smith Chart

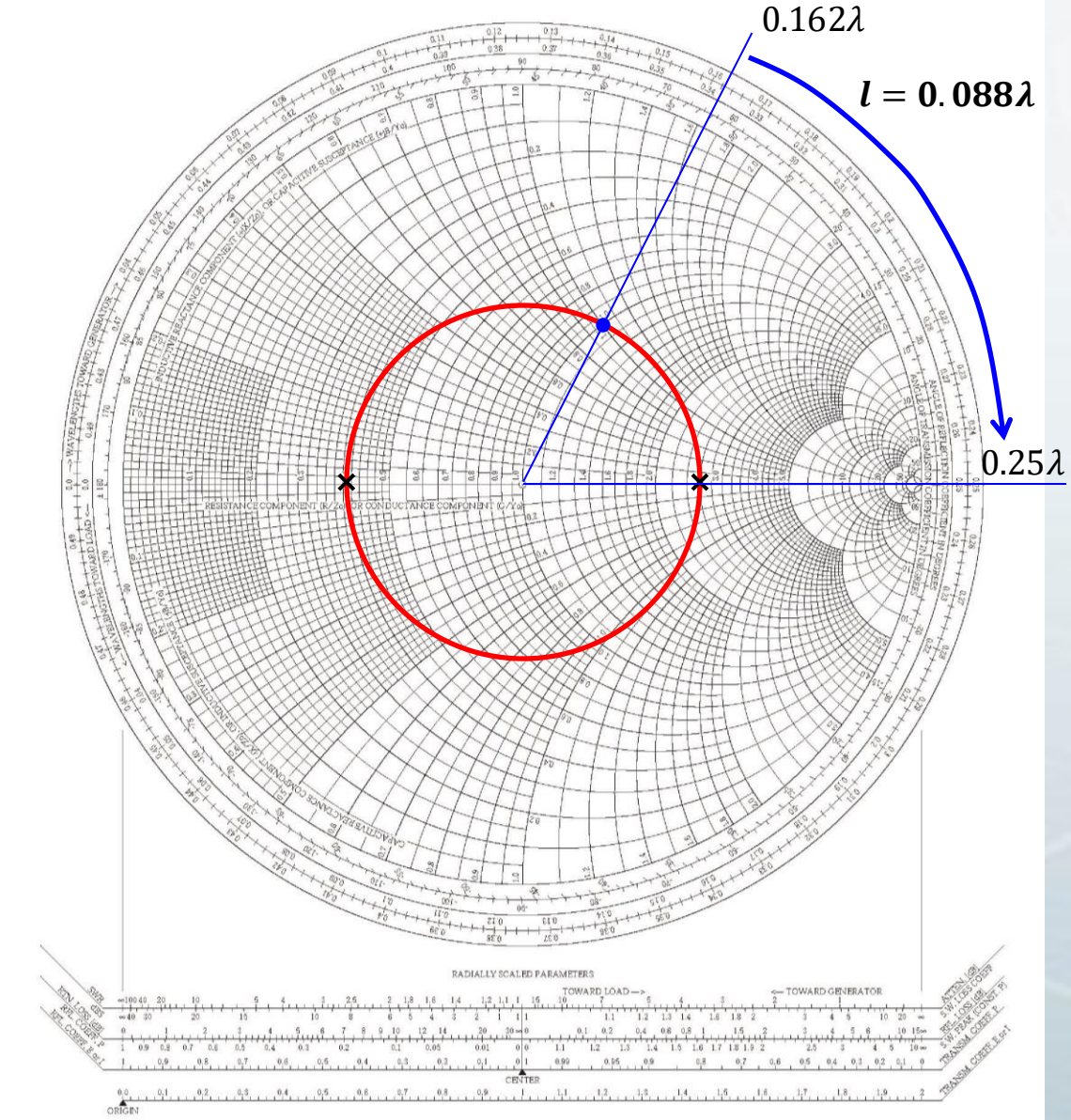


- What characteristic impedance Z_{OT} will match the input impedance of the line of length l ($Z_{in,l}$) to $Z_0 = 75\Omega$?

1. Read the normalized input resistance from the Smith chart and convert it to its actual input resistance.

$195\Omega = Z_{in,2}$

Smith Chart



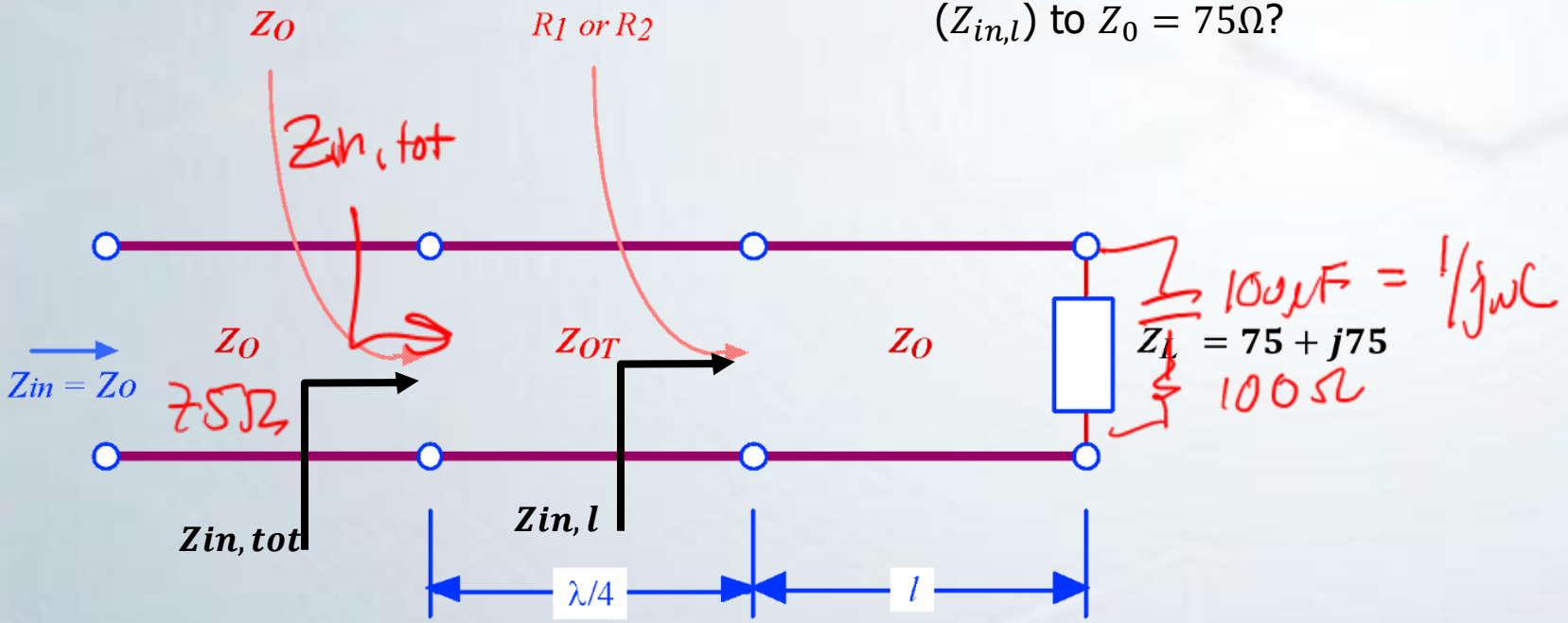
1. Read the normalized input resistance from the Smith chart and convert it to its actual input resistance.

$r = 2.6 \rightarrow$

$R = 2.6 \cdot 75\Omega = 195\Omega$

Smith Chart

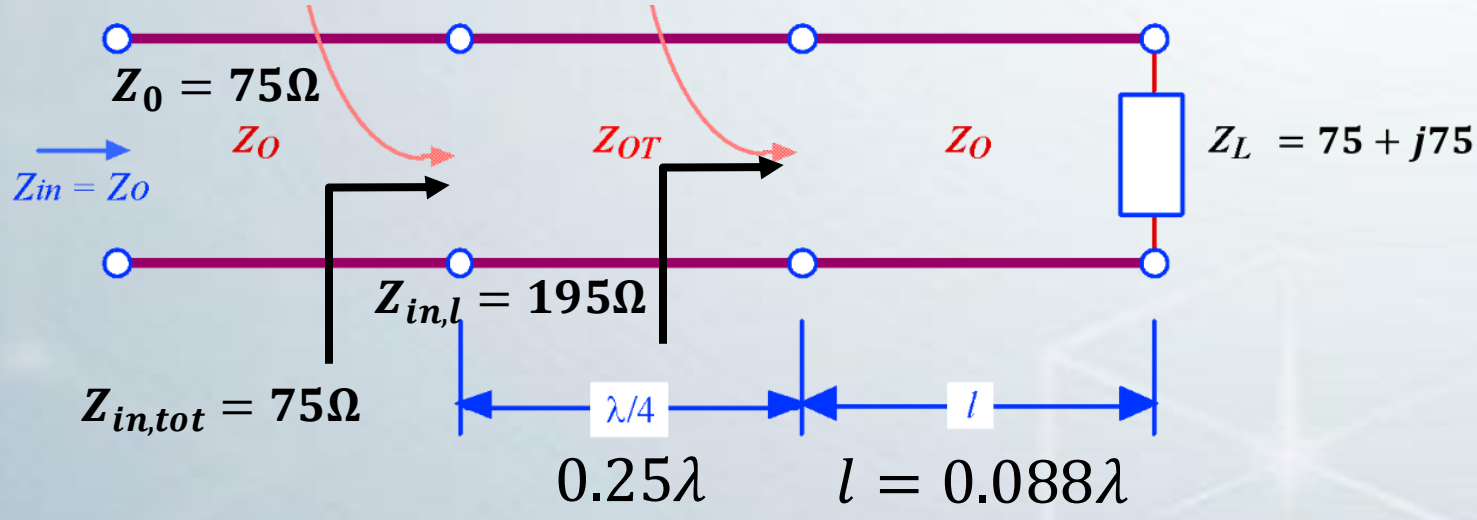
- What characteristic impedance Z_{OT} will match the input impedance of the line of length l ($Z_{in,l}$) to $Z_0 = 75\Omega$?



1. Read the normalized input resistance from the Smith chart and convert it to its actual input resistance.
2. Calculate Z_{OT} that is required to match $Z_{in,l}$ to $Z_0 = 75\Omega$

$$Z_{in,tot} = \frac{Z_{OT}^2}{Z_L} \rightarrow Z_{OT} = \sqrt{Z_{in,tot} \cdot Z_L} = \sqrt{75\Omega \cdot 195\Omega} = \mathbf{120.9\Omega}$$

Smith Chart



We have matched the feedline with $Z_0 = 75\Omega$ to a load impedance $Z_L = 75 + j75\Omega$!

Smith Chart

Summary

- The Smith Chart allows the graphical solution of the transmission line equation for *Z.*
- The Chart gives direct conversion between Γ and *Z.*

Wrap-Up

- Studio Session 2 due Friday (tomorrow)
- HW 2 due Monday